

Brayden Gess

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Active TS/SCI Clearance – CI Polygraph

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

B.S. Statistics and Machine Learning

GPA 3.5/4.0

Relevant Coursework: Introduction to Machine Learning, Applied Machine Learning, Principles of Imperative Programming, Data Structures and Algorithms, Statistical Computing, Statistical Inference, Modern Regression

EXPERIENCE

Data Engineer

Chantilly, VA

General Dynamics

Aug 2024 – Current

- Developed an agentic AI reasoning framework with CrewAI for autonomous LLM interaction with relational databases. Engineered agents with specialized domain knowledge and distinct personalities to generate queries, execute complex database operations and manage end-to-end workflows. Deployed MCP servers with custom and enterprise integrated toolsets, empowering agents to achieve sophisticated goals.
- Orchestrated scalable ETL pipelines using Airflow and PySpark for terabyte-scale datasets with optimized partitioning and joins, reducing latency by 25%. Supported real-time machine learning applications by delivering clean, high-throughput data to downstream models and teams.
- Deployed a semantic search system tailored for risk detection by integrating custom-trained embedding models into ElasticSearch. Tuned vector representations to capture language patterns for high-precision retrieval of threat indicators across communication metadata.
- Secured cloud translation workflows in the cloud using AWS S3 for persistent storage and automating API calls to accelerate multilingual processing.

INTERNSHIPS

Machine Learning Engineer Intern

Pittsburgh, PA

KeyLogic

May 2023 – Aug 2023

- Architected an unsupervised clustering system to classify wells, reducing input complexity to DOE's PARETO water-sharing network by 30%. Applied deep learning to improve forecasting of timelines and volumes
- Partnered with engineering teams to advance research on GENN pre-train derivatives for Carbon Capture Systems. Facilitating mathematical modeling techniques for model robustness and system performance

Data Science Intern

Pittsburgh, PA

Sheetz

May 2022 – Aug 2022

- Developed and deployed a machine learning model to optimize parking utilization across 650+ retail locations
- Achieved 6% increase in store profitability by reallocating underutilized space for higher margin assets
- Conducted traffic pattern analysis using the Databricks platform to accurately predict potential site profitability

PROJECTS/LEADERSHIP

Instructor – Advanced Catan Strategy [98-108]

Aug 2023 - Dec 2023

- Designed and taught an accredited, three-unit course on probabilistic strategy through the lens of *Catan*
- Explored advanced topics including Markov Chains, Bayesian reasoning and negotiation dynamics

Computer Vision Project [[opencv-mariokartwii](#)]

Aug 2023 – May 2024

- Applied video frame analysis and temporal smoothing for reliable event detection in gameplay footage
- Integrated Spotify API for dynamic soundtrack control based on real-time race state to enhance immersion

SKILLS

Machine Learning: Python, R, TensorFlow, Keras, PyTorch, Scikit-learn, Deep Learning, Model Deployment (MLflow)

Data Pipelines: PySpark, SQL, Airflow, Nifi, Databricks, S3, AWS Lambda, Hadoop, Kafka, Docker, Git, Linux